Transcendence of a Collatz-Related Constant via Sturmian Expansions

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Abstract

Let $C_{\text{Collatz}} = \frac{1}{2} - \sum_{n\geq 2} 2^{-\lfloor n \log_2 3 \rfloor} = 0.2862745023241087\ldots$, a constant arising as a natural density in the analysis of the accelerated Collatz map. We show that the binary expansion of C_{Collatz} is a Sturmian word, and deduce that C_{Collatz} is transcendental. The argument is not specific to the base 3: for every a with $\log_2 a$ irrational, the constant $C_a = 1 - \sum_{n\geq 1} 2^{-\lfloor n \log_2 a \rfloor}$ is transcendental. **Nothing here bears on the truth of the Collatz conjecture**; the result is a statement in transcendence theory that happens to concern a constant met in that context.

MSC: 11J81 (Transcendence), 68R15 (Combinatorics on words), 11B83.

1 The constant

For odd n, the accelerated Collatz step is $\text{step}(x) = (3x+1)/2^{v_2(3x+1)}$. Counting the odd integers whose first block does not immediately shrink leads to the density

$$C_{\text{Collatz}} \;=\; \tfrac{1}{2} - \sum_{n\geq 2} 2^{-\lfloor n \log_2 3 \rfloor}. \tag{1}$$

Summing (1) to $n = 300$ in 80-digit arithmetic gives

$$C_{\text{Collatz}} = 0.28627450232410874196\ldots$$

(the tail beyond $n = 300$ is below 2^{-475}). An equivalent closed form, $1 - \sum_{n\geq 1} 2^{-(n+\lfloor n \log_2(3/2) \rfloor)} = C_{\text{Collatz}}$, is machine-checked in Lean 4; it follows from the elementary identity $n + \lfloor n \log_2(3/2) \rfloor = \lfloor n \log_2 3 \rfloor$ (a one-line consequence of $\lfloor x - k \rfloor = \lfloor x \rfloor - k$ for $k \in \mathbb{Z}$).

2 The expansion is Sturmian

Write $\beta = \log_2 3 \in (1, 2)$, which is irrational, and let $B = (\lfloor n\beta \rfloor)_{n\geq 1}$ be the associated Beatty sequence. Define the gap word $w = (w_n)_{n\geq 1}$ by $w_n = \lfloor n\beta \rfloor - \lfloor (n-1)\beta \rfloor$.

Lemma 2.1 (Two-letter alphabet)**.** *$w_n \in \{1, 2\}$ for all n.*

Proof. $\lfloor x + \beta \rfloor - \lfloor x \rfloor \in \{\lfloor \beta \rfloor, \lfloor \beta \rfloor + 1\} = \{1, 2\}$ for any x, since $1 < \beta < 2$. □

Lemma 2.2 (Sturmian)**.** *w is a Sturmian word: its factor complexity satisfies $p(k) = k + 1$ for every $k \geq 1$.*

Proof. w is, by Lemma 2.1, the difference word of a Beatty sequence of irrational slope β; such words are exactly the mechanical words of slope $\beta - \lfloor \beta \rfloor$, which are Sturmian precisely because the slope is irrational (Morse–Hedlund [5]). □

Verification. Computing the factors of w directly for $n \le 4000$ gives $p(k) = k + 1$ for all $k \le 8$, with alphabet $\{1, 2\}$, as predicted.

Because the exponents $\lfloor n\beta \rfloor$ increase by $w_n \in \{1, 2\}$, the binary digits of the series in (1) are supported exactly on the Beatty set $\{\lfloor n\beta \rfloor\}$; the digit word of C_{Collatz} is therefore the characteristic word of that set, which is Sturmian by Lemma 2.2.

3 Transcendence

Theorem 3.1. *C_{Collatz} is transcendental.*

Proof. By Section 2 the binary expansion of C_{Collatz} is a Sturmian word, hence has factor complexity $p(k) = k + 1$, which is $o(k \log k)$ and in particular sub-linear in the sense required. A real number whose base-b expansion is Sturmian is transcendental: this is the theorem of Ferenczi–Mauduit [1], and follows also from the complexity criterion of Adamczewski–Bugeaud [2], since an algebraic irrational cannot have an expansion of complexity $p(k) = k + 1$. As C_{Collatz} is irrational (its expansion is aperiodic, being Sturmian), it is transcendental. □

Corollary 3.2. *For every real $a > 1$ with $\log_2 a \notin \mathbb{Q}$, the constant $C_a = 1 - \sum_{n \ge 1} 2^{-\lfloor n \log_2 a \rfloor}$ is transcendental. In particular this holds for every integer a that is not a power of 2.*

Proof. Identical, with $\beta = \log_2 a$; only irrationality of β is used. If $1 < \beta < 2$ fails, replace w by the gap word of $\lfloor n\beta \rfloor$, whose alphabet is $\{\lfloor \beta \rfloor, \lfloor \beta \rfloor + 1\}$; the mechanical-word argument is unchanged. □

Remark 3.3. We checked C_{Collatz} against the CODATA physical constants and the standard mathematical constants and found no match to within 10^{-3}; we know of no closed form for it, and it is transcendental by Theorem 3.1. It is not, however, a constant of a new kind: it belongs to the classical family $\sum_{n \ge 1} b^{-\lfloor n\alpha \rfloor}$, studied since Böhmer [6], who proved transcendence when α has unbounded partial quotients, and later by Danilov [7], Davison [8] and Adams–Davison [9], the last of whom give the continued fraction of $(b - 1) \sum_{n \ge 1} b^{-\lfloor n\alpha \rfloor}$ explicitly for every irrational α. What is specific here is only that this member of the family, with $b = 2$ and $\alpha = \log_2 3$, is the one the Collatz observable produces.

Remark 3.4 (Diophantine flavour)**.** The continued fraction of C_{Collatz} begins

$$[0;\, 3,\, 2,\, 36,\, 2056,\, 70368878395648,\, \ldots];$$

the very large early partial quotients reflect exceptionally good dyadic-rational approximations. Only the terms stable across two independent precisions are reported.

Reproducibility

The value, the gap alphabet and the factor complexity are recomputed by `tools/verify_for_papers.py`; the closed-form identity is checked by Lean 4 (`blockSum_eq_cCollatzClosed`, no custom axioms).

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References

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